We have conducted ablation experiments to empirically validate the role of each core component. Specifically, the proposed PGAG framework is decomposed into three fundamental drivers: Graph Consistency (GC), Structure Consistency (SC), and Change Feedback (CF). To ensure each component functions independently, the experiments are configured with the following unified settings: patch size μ = 5, step Δ = 1, and adjacency ratio Ar = 0.01. Based on this configuration, ablation experiments are performed on the Yellow River dataset, and the results are exhibited as follows.

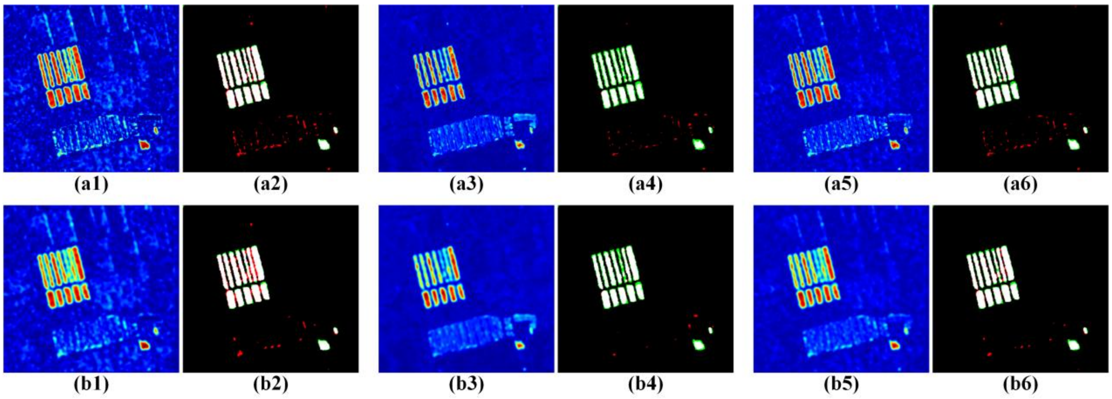

Fig 1. Ablation experiments results of the proposed PGAG method on Yellow river dataset. (a1) DI of GC without CF, (a2) Change map of GC without CF, (a3) DI of SC without CF, (a4) Change map of SC without CF, (a5) DI of PGAG without CF, (a6) Change map of PGAG without CF, (b1) DI of GC with CF, (b2) Change map of GC with CF, (b3) DI of SC with CF, (b4) Change map of SC with CF, (b5) DI of PGAG, (b6) Change map of PGAG.

TABLE I

QUANTITATIVE RESULTS OF ABLATION EXPERIMENT

Method	FN	FP	OA	F1	κ
GC without CF	477	800	0.9857	0.8824	0.8748
SC without CF	1692	211	0.9786	0.7899	0.7790
PGAG without CF	942	380	0.9852	0.8675	0.8597
GC with CF	375	864	0.9861	0.8877	0.8803
SC with CF	1167	110	0.9857	0.8653	0.8579
PGAG	635	339	0.9891	0.9049	0.8991

The empirical evidence, summarized in the provided results, reveals distinct functions for each component. The Change Feedback mechanism operates primarily as a regularization or smoothing operator, refining the detection map and contributing an average improvement of 4.13% in the κ coefficient, as evidenced by comparing results with and without CF. Graph Consistency (GC) is primarily focused on change enhancement, effectively highlighting potential changed regions. This leads to a lower missed detection rate but a relatively higher false alarm rate. Its integration yields a substantial average gain of 6.10% in κ. Conversely, Structure Consistency (SC) emphasizes spatial smoothness and homogeneity, resulting in fewer false alarms at the cost of potentially missing some changes, with a corresponding modest average improvement of 0.18% in κ. Based on this quantitative analysis, we conclude that Graph Consistency (GC) provides the most critical contribution to the final detection accuracy within the proposed PGAG framework.